

```

1  #include <DHT.h>
2  #define DHTPIN 2
3  #define DHTTYPE DHT22
4  DHT dht(DHTPIN, DHTTYPE);
5
6  int relayPin = 7; // Connected to the relay module
7  float humidityThreshold = 55.0; // Set required humidity level
8
9  void setup() {
10     Serial.begin(9600);
11     dht.begin();
12     pinMode(relayPin, OUTPUT);
13     digitalWrite(relayPin, HIGH); // Keep relay off initially
14 }
15
16 void loop() {
17     float humidity = dht.readHumidity();
18     Serial.print("Humidity: ");
19     Serial.println(humidity);
20
21     if (humidity < humidityThreshold) {
22         digitalWrite(relayPin, LOW); // Turn on humidifier
23     } else {
24         digitalWrite(relayPin, HIGH); // Turn off humidifier
25     }
26
27     delay(5000); // Wait 5 seconds before next reading
28 }
29
30

```

Fig. 3: Snippet of the source code

the ultrasonic humidifier is connected to the relay's normally open (NO) pin.

The DHT22 sensor reads the humidity, and the Arduino processes this data by comparing it to a preset humidity threshold. If the humidity falls below the threshold, the relay activates the mist maker. Once the desired humidity level is reached, the relay turns off the humidifier.

The entire system shares the same 5V power rail and can be powered using a regulated 5V power supply, such as a phone charger adaptor or a USB 5V power bank.

Note. An LCD or OLED display can be added to monitor real-time

humidity levels after modifying the source code and circuit diagram.

Software

This system is based on the Arduino IDE platform. The code used here is straightforward: the sensor continuously monitors the environmental humidity, and if the humidity drops below the set threshold, the humidifier is automatically turned on. It can also be configured to turn

off the humidifier if the humidity becomes too high.

Construction and testing

First, upload the source code to the Arduino Uno. Then, connect all the components on the breadboard as shown in the circuit diagram (refer Fig. 2).

For testing, power on the system. Based on the humidity level, the automatic operation of the humidifier can be observed. When the humidity drops below the set threshold, the humidifier turns on automatically.

Since the humidifier is connected to the relay contacts, it turns on or off depending on whether the relay is energised or de-energised. The relay is controlled through pin D7 of the Arduino Uno. **EFY**

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Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

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